

COURSE TITLE : BIOMEMS AND NANO TECHNOLOGY
COURSE CODE : 6244
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPICS	PERIODS
1	Microsystems and Nanotechnology	15
2	MEMS materials and Nanostructure synthesis	15
3	MEMS devices	15
4	Nanosensors and nanodevices	15
TOTAL		60

Course General Outcome :

MODULE	G.O	ON COMPLETION OF THE STUDY OF THIS MODULE THE STUDENT WILL BE ABLE:
1	1	To understand microsystems
	2	To understand Nanotechnology
2	3	To understand MEMS materials
	4	To understand nanostructure synthesis
3	5	To understand MEMS devices
4	6	To understand Nanosensors and nanodevices for clinical diagnostics

G.O.: General Outcome

On completion of the study the student will be able:

MODULE I MICROSYSTEMS AND NANOTECHNOLOGY

1.1.0 To understand microsystems

- 1.1.1 To state the concept of Microsystems
- 1.1.2 To define MEMS and Bio MEMS
- 1.1.3 To explain the evolution of Microfabrication

1.2.0 To understand Nanotechnology

- 1.2.1 To explain Nanotechnology
- 1.2.2 To describe Silicon microfabrication techniques
- 1.2.3 To define photolithography (high resolution), Ion Implantation, oxidation, diffusion, sputtering, epitaxial growth and etching
- 1.2.4 To explain the design of flow processes in bulk manufacturing
- 1.2.5 To define surface micro machining
- 1.2.6 To describe the LIGA process
- 1.2.7 To describe EFAB fabrication

- 1.2.8 To describe Micro system packaging

MODULE II MEMS MATERIALS AND NANOSTRUCTURE SYNTHESIS

2.1.0 To understand MEMS materials

- 2.1.1 To list MEMS materials
- 2.1.2 To explain common Bio MEMS polymers
- 2.1.3 To define micro fluidics
- 2.1.4 To define micro arrays
- 2.1.5 To explain Polymerase chain reaction (PCR)
- 2.1.6 To list the elements required for PCR

2.2.0 To understand nanostructure synthesis

- 2.2.1 To describe functional polymers and dendrimers
- 2.2.2 To explain Microelectronic Array Devices
- 2.2.3 To state the concept of DNA diagnostics
- 2.2.4 To explain nanofabrication applications
- 2.2.5 To explain nanotechnology manufacturing

MODULE III MEMS DEVICES

3.1.0 To understand MEMS devices

- 3.1.1 To list MEMS devices
- 3.1.2 To explain Pressure sensors
- 3.1.3 To explain accelerometers
- 3.1.4 To explain micromotors
- 3.1.5 To explain micropumps
- 3.1.6 To explain microvalves
- 3.1.7 To explain thermal sensors and actuators
- 3.1.8 To explain prosthetics made of MEMS

MODULE IV NANOSENSORS AND NANODEVICES

4.1.0 To understand Nanosensors and nanodevices for clinical diagnostics

- 4.1.1 To explain nanostructures for drug delivery
- 4.1.2 To explain nano arrays
- 4.1.3 To describe the use of nano analytical devices and systems
- 4.1.4 To explain the potential use of DNA and other biomolecules for computing
- 4.1.5 To explain the potential use of DNA and other biomolecules for ultra high density data storage
- 4.1.6 To describe the Application of Nanotechnology to Medical Therapy

CONTENTS

MODULE I MICROSYSTEMS AND NANOTECHNOLOGY

Introduction to Microsystems - MEMS and BioMEMS - Evolution of Microfabrication - Introduction to Nanotechnology – Definition of Silicon microfabrication techniques such as photolithography (high resolution), Ion Implantation, oxidation, diffusion, sputtering, epitaxial growth and etching-Design of flow processes in bulk manufacturing-surface micro machining- LIGA process –EFAB fabrication-Micro system packaging

MODULE II MEMS MATERIALS AND NANOSTRUCTURE SYNTHESIS

MEMS materials - Polymer materials -common Bio MEMS polymers-micro fluidics-micro arrays Polymerase Chain Reaction (PCR)-elements of PCR.

Nanostructure synthesis-functional polymers and Dendrimers-Microelectronic Array Devices- DNA Diagnostics and nanofabrication applications-Nanotechnology Manufacturing

MODULE III MEMS DEVICES

MEMS Devices - Pressure sensors, accelerometers, micromotors, micropumps, microvalves, thermal sensors and actuators- prosthetics made of MEMS.

MODULE IV NANOSENSORS AND NANODEVICES

Nanosensors and nanodevices for clinical diagnostics –nanostructures for drug delivery - nano arrays - use of nano analytical devices and systems –potential use of DNA and other biomolecules for computing and ultra high density data storage - Application of Nanotechnology to Medical Therapy.

TEXT BOOKS

1. BioMEMS and Biomedical Nanotechnology - Ferrari , Mauro Springer 2006
2. Fundamentals of BioMEMS and Medical Microdevices - Steven S. Saliterman, SPIE Press Monograph, 2006

REFERENCES

1. MEMS&Microsystems - Tai-Ran Hsu, TMH, New Delhi
2. Microsystem Design - S.Senturia,, Kluwer Academic Press, 2000
3. Micromachined Transducers Source book - G.Kovacs, McGraw Hill, NY, 1998
4. The Science and Engineering of Microelectronic Fabrication - S.A.Campbell, Oxford University Press, 1996
5. Nanotechnology: A Gentle Introduction to the Next Big Idea - Mark A Ratner, Daniel Ratner PHI
6. MEMS and NEMS: Systems, Devices and Structures - Sergey E. Lyshevski, CRC Press, 2002